THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET B

Chapter 5: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Is 7314 divisible by 3? 1

(a) Only if the last digit is even **(b)** Only if the number is prime
(c) Yes **(d)** No

2. Is 6912 divisible by 9? 1

(a) Yes **(b)** Only if the number is prime
(c) Only if the last digit is even **(d)** No

3. Is 7062 divisible by 11? 1

(a) No **(b)** Only if it is odd
(c) Only if it ends in 1 **(d)** Yes

4. The smallest digit that replaces the blank in 81_4 so that the number is divisible by 3 is: 1

(a) 3 **(b)** 2 **(c)** 7 **(d)** 4

5. The LCM of 8 and 20 is: 1

(a) 48 **(b)** 160 **(c)** 4 **(d)** 40

6. The LCM of 9 and 12 is: 1

(a) 45 **(b)** 108 **(c)** 3 **(d)** 36

7. A number is divisible by 6 if it is divisible by: 1

(a) both 2 and 3 **(b)** 5 and 2 **(c)** 2 and 4 **(d)** 3 and 9

8. The Kaprekar constant for four-digit numbers is: 1

(a) 1729 **(b)** 1234 **(c)** 6174 **(d)** 1089

9. The smallest prime number is: 1

(a) 2 **(b)** 0 **(c)** 1 **(d)** 3

10. The digit sum of 9999 is: 1

(a) 18 **(b)** 9 **(c)** 27 **(d)** 36

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What are co-prime numbers? Give an example. 2

12. Find the HCF of 18 and 48. 2

13. Find the LCM of 6, 8 and 12. 2

14. What is a palindromic number? Give two examples. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Find the smallest number divisible by 12, 18 and 30. 3

16. Find the smallest number divisible by 8, 12 and 20. 3

17. Find the smallest number divisible by 15, 20 and 25. 3

OR

Find the smallest number divisible by 6, 9 and 15. 3

18. Find the smallest number divisible by 14, 21 and 35. 3

OR

Find the smallest number divisible by 10, 16 and 24. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Apply Kaprekar's routine to 954. 5
(ii) What does the sequence show?
(iii) Try 100 and say what happens.

OR

(i) Find the smallest 4-digit number divisible by 9 and 11. 5
(ii) Find the largest 4-digit number divisible by both.

20. **Case study:** A teacher wants to arrange 144 red and 180 blue flags in equal rows so that each row has flags of one colour only and rows have equal flags. 5
(i) What is the largest number of flags per row?
(ii) How many rows of each colour?
(iii) Why HCF?

OR

(i) Show that a number is divisible by 9 iff its digit sum is divisible by 9. 5
(ii) Test 70524.

END OF QUESTION PAPER — Total: 40 Marks

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